

Helpful Linux Commands

File & Directory Management

Command	Description
ls	Lists the contents of the current directory.
pwd	Prints the working directory (shows your current location).
cd <path>	Changes directory. Use cd .. to move up one level.
cat <file>	Displays the entire content of a file to the screen.
nano <file>	Opens the file in the nano text editor for editing.
rm <file>	Removes (deletes) a file.
mv <source> <dest>	Moves or renames a file or directory.

grep <pattern> <file>	Searches for lines matching a pattern within files. Often used with pipes (`
sed -n 'A,Bp' <file>	Prints a specified range of lines (A to B) from a file.

System, Networking, & Process Management

Command	Description
sudo netstat -tlnp grep <port>	Displays listening ports (-tln) and the associated process ID (-p). Used to check if a port (like 8000 or 5000) is in use.
kill -9 <PID>	Forcibly terminates a process using its Process ID (PID), which is found via netstat or ss.
curl <URL>	Transfers data from a URL; used to test if a local web service is running (e.g., curl http://localhost:8000).
python3 -m http.server 8000	Starts a basic, temporary Python web server on port 8000 in the current directory.
which <command>	Shows the full path to an executable command.

clear	Clears the terminal screen.
history	Displays a list of previously executed commands.

Google Cloud & ADK (Agent Development Kit)

Command	Context	Description
gcloud auth login	Authentication	Logs in to Google Cloud, granting access to services.
gcloud auth application-default login	Authentication	Logs in for Application Default Credentials (ADC), used by many Google Cloud libraries/SDKs.
gcloud compute firewall-rules list	Networking	Lists all VPC firewall rules, often piped with grep to filter results.
gcloud compute instances describe <instance>	Compute Engine	Retrieves detailed information about a virtual machine instance.
gcloud alloydb clusters describe <cluster>	Databases	Retrieves details about an AlloyDB cluster.

gcloud services enable <service>	APIs	Enables a specific Google Cloud API service for the project.
export <VAR>=<value>	Environment	Sets a session-specific environment variable (e.g., project ID, region, or credentials path).
adk web --host 0.0.0.0 --port 8000	ADK/Server	Starts the ADK web server on the specified host and port for agent interaction.
./toolbox --tools-file "tools.yaml"	ADK/Tools	Executes the local toolbox application, typically to start a tool service for a Gemini Agent using a tools.yaml configuration file.
source .venv/bin/activate	Python	Activates the Python virtual environment named .venv.
python <script.py>	Python/Agent	Runs a Python agent script, passing an input string.

Database (PostgreSQL/AlloyDB)

Command	Description
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psql -h <IP> -p 5432 -U postgres -d postgres	Connects to a PostgreSQL database (like AlloyDB) using the host IP, port, username, and database name.
pg_dump ... > full_dump	Creates a backup/dump of the PostgreSQL database content.
sudo apt-get install -y postgresql-client	Installs the necessary client tools to interact with a PostgreSQL database.